5.3 1(4) 2(2) 6(1)

#10 (4)

$x = e^t + e^{-t}$, $y = e^t - e^{-t}$; $t = 0$

$x' = e^t - e^{-t}$

$x'' = e^t + e^{-t}$

$y' = e^t + e^{-t}$

$y'' = e^t - e^{-t}$

$|x'y'' - x''y'| = (e^t - e^{-t})^2 - (e^t + e^{-t})^2$

$= -4$

$(x')^2 + (y')^2 = (e^t - e^{-t})^2 + (e^t + e^{-t})^2$

$= 2e^{2t} + 2e^{-2t}$

$\therefore$ $t=0$ 에서 곡률은

$K = \dfrac{|x'y'' - x''y'|}{((x')^2 + (y')^2)^{\frac{3}{2}}} = \dfrac{4}{4^{\frac{3}{2}}} = \dfrac{4}{8}$

$= \dfrac{1}{2}$

반지름 $= 2$

#20 (2)

$x = 2t - 3$, $y = t^2 - 1$

$x' = 2$

$x'' = 0$

$y' = 2t$

$y'' = 2$

$x'y'' - x''y' = 4$

$(x')^2 + (y')^2 = 4 + 4t^2$

$K = \dfrac{4}{(4 + 4t^2)^{\frac{3}{2}}}$

$\therefore (4 + 4t^2)^{\frac{3}{2}}$: 최소 $\therefore t = 0$

$\therefore (-3, -1)$ 에서 K 최대

그때의 곡률 반지름은 2